Ground-Level Ozone Forecasting Using Explainable Machine Learning

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Abstract. The ozone concentration at ground level is a pivotal indicator of air quality, as elevated ozone levels can lead to adverse effects on the environment. In this study various machine learning models for ground-level ozone forecasting are optimised using a Bayesian technique. Predictions are obtained 24 h in advance using historical ozone data and related environmental variables, including meteorological measurements and other air quality indicators. The results indicated that the Extra Trees model emerges as the optimal solution, showcasing competitive performance alongside reasonable training times. Furthermore, an explainable artificial intelligence technique is applied to enhance the interpretability of model predictions, providing insights into the contribution of input features to the predictions computed by the model. The features identified as important, namely PM_{10}, air temperature and CO_2 concentration, are validated as key factors in the literature to forecast ground-level ozone concentration.

Keywords: Ozone concentration · Time series forecasting · Explainable artificial intelligence

1 Introduction

Air quality is a crucial aspect of environmental health, with atmospheric composition significantly impacting ecosystems and organisms. Ozone (O_3) plays a dual role as both a protective layer in the stratosphere and a potential hazard at ground level. While the ozone layer shields against harmful UV radiation, elevated concentrations of ground-level ozone can negatively affect human health, ecosystems, and agricultural output. This poses a significant challenge to air quality, particularly in urban areas with heavy traffic [1]. As concerns about atmospheric pollution and its adverse effects intensify, it has become imperative to develop precise and effective methods for accurate predictions of O_3 concentration in the air, as they enable proactive management strategies to mitigate pollution, such as coordinating traffic patterns. In this context, Machine Learning (ML) and Deep Learning (DL) techniques have emerged as a promising

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tool to improve prediction accuracy. However, environmental experts hesitate to fully embrace these techniques in real-world applications and policymaking due to their opaque nature.

This study aims to explore and assess the efficacy and efficiency of different ML models in predicting future ground-level O_3 concentration based on air quality and meteorological measurements over the past hours. Then, the model obtaining the best results is explained through an eXplainable Artificial Intelligence (XAI) technique based on quantitative explanation rules, denoted as RULEx, thus obtaining an accurate and interpretable framework for forecasting concentrations O_3.

The remainder of this paper is structured as follows. First, Sect. 2 summarises contributions related to O_3 forecasting found in the literature. Second, Sect. 3 introduces the proposed methodology, and third, Sect. 4 presents the results that have been obtained. Finally, Sect. 5 concludes the paper and leads to future work lines.

2 Related Work

Several approaches in the literature explored the integration of ML solutions for the forecasting of O_3 in the context of air pollution, with initiatives to improve prediction models, thus considering the complex interrelationships between air quality and meteorological elements. First, the authors in [2] conducted a comprehensive review focused on presenting the latest advances in the area, emphasising the nature of the models utilized and the input variables employed as input data. The model repertoire comprised Support Vector Machines (SVM), Artificial Neural Networks (ANN), decision trees, and hybrid models. The findings indicated that while SVM-based models outperformed others in accuracy, ANN are extensively employed in existing literature. Additionally, the analysis revealed that nitrogen oxides (NO_2, NO), particulate matter (PM_{10}, $PM_{2.5}$), and ozone (O_3) are the most frequently examined in the reviewed articles. Furthermore, in the paper presented in [3], the authors used meteorological data from Kennewick (USA) as input data to a hybrid approach combining different models, namely random forest and linear regression. The results showcased the system's reliability with significantly reduced computational resources. In this way, the benefits of introducing not only air pollution observances but also exogenous variables such as temperature or wind speed are highlighted in [4]. The application is carried out on a six-year data set of hourly O_3 concentrations and meteorological measurements from Béthune (France). The models were trained with only air observations and with these measurements combined with exogenous data, and the models trained with exogenous variables showed the best predictive power.

Additionally, Neural Networks (NN) are extensively utilized in various applications in the literature. For example, the authors of the article in [5] proposed an integrated model based on multilayer perceptron, combining available data from other environmental models and measurements on-site of meteorological and

air pollution parameters in Slovenia. Validation results demonstrated that integrated models significantly improve O_3 forecasts and contribute to more effective alert systems. Moreover, a novel approach utilizing a spatiotemporal graph NN to predict O_3 concentration dynamics in urban areas, using meteorological and air quality data from Houston, Texas, was introduced in [7]. The research incorporated various configurations of time lags, and three forecast horizons (1, 3, 6 h ahead). The results demonstrated the efficacy of the proposed model in recognising intricate spatio-temporal patterns within the data. Another example is found in [6], where the authors introduced a comparative study between a hybrid nonlinear regression (NLR) model and a NN model, both tailored to predict next-day maximum 1-hour average ground-level O_3 concentrations in Louisville, USA, during 1998 and 1999. The NLR model demonstrated a slight superiority over the NN model using observed meteorological data. Furthermore, meteorological data and air pollutant observations were used as input to a DL model with long short-term memory (LSTM) layers in [8]. Predictions of O_3 concentrations are obtained for the next 48 h, thus improving both temporal and spatial air quality.

Finally, in terms of an XAI perspective, the tool denoted eXplaining Air Quality (AQX) was presented in [9] as a visual analytics system developed to assist experts in validating DL models with domain knowledge. Thus, multiple coordinated views were used to present input feature contributions across temporal and spatial dimensions. The methodology was tested using a case study in the field of O_3 forecasting, showing the effectiveness and utility of the tool. Furthermore, another DL approach has been developed in the domain of forecasting evapotranspiration [10], an ecological concern similar to the forecasting problem O_3. The authors optimized and trained a LSTM model, and explanations were obtained with SHapley Additive exPlanations (SHAP), aligning with existing literature on the primary significant exogenous features.

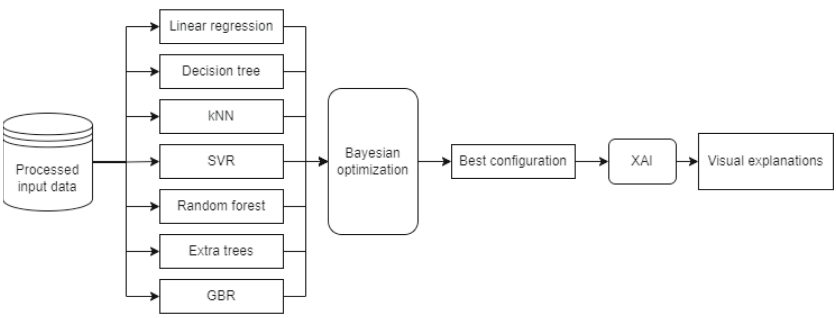


Fig. 1. Methodology workflow.

3 Materials and Methods

This paper introduces the application of a suite of ML models designed for O_3 forecasting, followed by an explanation of the best-performing model using an XAI technique. The overall methodology is depicted in Fig. 1, with each step elaborated in subsequent sections.

3.1 Machine Learning

ML models have consistently shown strong performance in the prediction of O_3 concentration. To identify the most effective option, various techniques, including Linear Regression (LR), Decision Trees (DT), k nearest neighbours (kNN), Support Vector Regression (SVR), Random Forest (RF), Extra Trees (ET), and Gradient Boosting Regression (GBR), are optimized and evaluated. The models are delineated as follows.

- **LR** models the relationship between the dependent variable and one or more independent variables by fitting a linear equation.
- **DT** divides the feature space into a set of regions and assigns a simple model to each region, making predictions based on the region the input falls into.
- **kNN** makes predictions by averaging the target values of the k-nearest data points in the feature space.
- **SVR** is a variant of the regression-adapted support vector machines.
- **RF** is an ensemble learning method that constructs a multitude of decision trees and outputs the mean prediction of each individual tree.
- **ET** is an extension of the Random Forest method, thus building decision trees from random training data samples, making it easier to train.
- **GBR** builds an ensemble of weak learners (typically decision trees) sequentially, each new model correcting the errors made by the previous ones, using a gradient descent method to minimize a loss function during training.

The stochastic approaches, including those based on trees, namely DT, RF, ET, and GBR, are executed 25 times to ensure robustness and reliability of the results.

3.2 Hyperparameter Optimization

The ML models underwent a process of optimization using a Bayesian technique to iteratively refine the hyperparameters governing the ML models' architecture and training process. This technique employs Bayesian inference to construct a probabilistic model of the objective function and utilizes this model to determine the most promising points. Optimization balances exploring uncertain regions with exploiting areas likely to yield improvement, ultimately converging toward the optimal solution. Then, the hyperparameters optimized and their range are detailed in Table 1. LR model has no parameters to optimize, and it serves as a baseline model in this study. Furthermore, optimization encompasses the input data window for all the models, including LR, thus refining the timeframe of the input window, and, as the data is hourly, this entails testing window sizes ranging from 24 to 72 h, covering every hourly increment in between.

Table 1. Hyperparameters ranges for each model optimization.

Model	Hyperparameters	Range
DT	max_depth	[2, 50]
kNN	k	[2, 7]
SVR	C	[0.0001, 5]
	epsilon	[0.0001, 1]
RF	max_depth	[2, 50]
	n_estimators	[10, 256]
ET	max_depth	[2, 50]
	n_estimators	[10, 256]
GBR	max_depth	[2, 50]
	n_estimators	[10, 256]

3.3 XAI

The XAI method used in this experiment is denoted as RULEx, presented in [11], and based on quantitative association rules (QARs), obtained with MOQAR method. MOQAR [12] is a multi-objective evolutionary algorithm applied for discovering QARs, to identify optimal configurations, with an approach specifically suitable for time series data because it can handle association rules in continuous intervals, thus allowing the capture of temporal relationships between events.

In this way, the RULEx technique aims to enhance the interpretability of time series forecasting through the visual representations of sets of rules, thus referring to patterns that capture relationships among variables. RULEx is a model-agnostic explainability technique, thus elucidating how the model works by illustrating the most important values for the time series forecasting task once the predictions have been obtained. The output is a chart-based representation colorized to reflect the significance of the input values in influencing the predictions for each distinct prediction horizon. RULEx has been recently applied to obtain time series explainability in [13].

4 Results

This section presents the results derived from this study. The experimental process involves fine-tuning the models delineated earlier. After that, the predictions generated by the optimal model are explained by employing the RULEx technique to offer interpretable explanations of the model's predictions.

4.1 Input Data

The input data set comprises a collection of hourly measurements that include air quality and meteorological data from 2006 to 2023. These measurements were

recorded at a monitoring station located in the metropolitan area of Seville, Spain, at the coordinates 42.8748° N and 1.6256° W. The meteorological conditions included the average temperature, wind direction (DD), and wind speed (VV), and the air quality factors considered were carbon monoxide (CO), nitrogen dioxide (NO_2), and particles of 10 µm or less (PM_{10}). For more information about the data, [14] can be consulted. The O_3 measurements were also obtained using the reference monitoring method [15]. The model is trained with data from 2006 to 2021, hyperparameters are validated using data from 2022, and model performance is assessed on data from 2023. Furthermore, in this work, air quality measures and meteorological variables are used to forecast O_3 conditions for the next 24 h, equivalent to one day ahead. This approach incorporates historical data to predict future O_3 levels, enabling more accurate and timely forecasts.

4.2 Evaluation Metrics

The models' evaluation is conducted using metrics that are widely utilized in forecasting tasks, thus ensuring robustness and comparability across different methodologies, namely Mean Absolute Error (MAE), in Eq. (1), Root Mean Squared Error (RMSE), in Eq. (2), and R-squared (R2), in Eq. (3). MAE and RMSE metrics quantify the disparities between the predictions and the actual values, ranging from 0 to infinity depending on the data, and values closer to 0 indicate higher predictive precision. On the contrary, R2 ranges from 0 to 1, where 1 denotes the complete explanation of the dependent variable. The training time of each model is also measured on a machine with the following specifications: 8 GB RAM and 512 GB SSD disk. The mathematical expressions of the quality measures are described below, where y refers to actual data, $\hat{y}$ to predicted data, n is the total number of samples, and $\bar{y}$ is the mean value of the input samples.

$$MAE = \sum_{i=1}^{n} |y_i - \hat{y}_i| \tag{1}$$

$$RMSE = \sqrt{\frac{1}{n} \Sigma_{i=1}^{n} \left(y_i - \hat{y}_i\right)^2} \tag{2}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2} \tag{3}$$

4.3 ML Optimization

In this section, the results of the Bayesian optimization process are introduced and compared using the evaluation metrics presented above.

The best parameters obtained for each model are presented in Table 2, and Table 3 outlines the error metrics and training times associated with each optimal configuration, ordered by RMSE. The variation in window size for each model

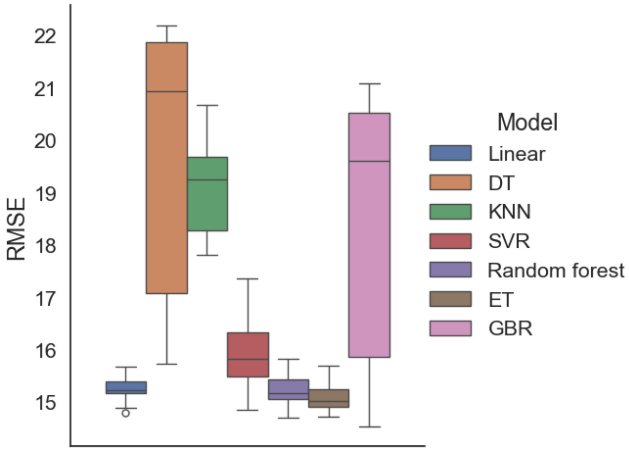


Fig. 2. RMSE boxplot for the Bayesian optimization process.

Table 2. Best hyperparameters configuration for each model.

Model	Window size	Hyperparameters
DT	44	{max_depth: 6}
kNN	38	{k: 6}
SVR	62	{C: 0.7020, epsilon: 0.1982}
RF	39	{max_depth: 27, n_estimators: 180}
ET	39	{max_depth: 27, n_estimators: 180}
GBR	66	{max_depth: 3, n_estimators: 180}

Table 3. Error metrics for the best configuration of each model.

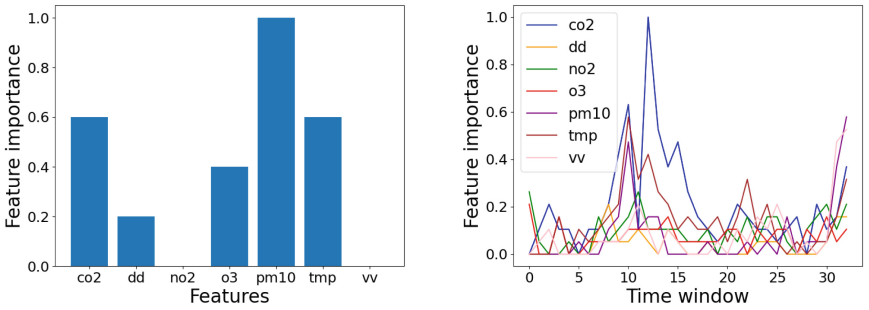
Model	MAE	RMSE	R2	Time (s)
GBR	**11.38**	**14.55**	**0.60**	416.79
ET	11.57	14.73	**0.60**	52.41
LR	11.52	14.80	0.59	0.09
RF	11.75	14.88	0.59	186.14
DT	12.33	15.73	0.54	0.91
SVR	12.26	15.44	0.57	81.52
KNN	14.22	17.81	0.46	**0.01**

may stem from the distinct sensitivity of each algorithm to different lengths of window, as certain models need longer windows to capture patterns in the data. Then, Fig. 2 illustrates the variations of RMSE metric across all the different configurations of each model tested during the optimization phase.

Based on Table 3, the GBR model has the best performance in terms of evaluation metrics, however, it requires substantially more time than the other models. In addition, concerning Fig. 2, the mean RMSE for different configurations of GBR models is considerably higher, exhibiting a wide range of variation. With this concern, the Extra Trees (ET) is selected as the optimal option due to its performance comparable to the best GBR configuration in terms of error metrics, with slightly higher values for MAE and RMSE and the same R2 value, and with a significantly shorter training time of 52.41 s. Moreover, all the tried ET models collectively perform better, as Fig. 2 shows. Finally, regarding the input time window, most models demonstrated improved performance with approximately two days of input data, given the hourly data frequency. Notably, the ET optimal performance is achieved when trained on the preceding 39 h of data.

4.4 XAI

The selected model from the previous section, namely ET with a maximum depth of 27, 180 estimators, and 39 items as input time window, is elucidated through RULEx. A set of QARs is generated independently for each prediction horizon, utilizing both the real input data (each variable in each time window) and the predictions obtained by the ET model.



(a) Global influence of each feature.

(b) Feature importance across time windows.

Fig. 3. XAI outputs.

Figure 3a summarizes the importance of each feature in determining the forecasts. The most significant feature discovered is PM_{10}, the variable that indicates the presence of particles smaller than 10 µm, and the past values of O_3 are the fourth feature in importance. These results align with the findings presented in the review paper in [2], whose analysis revealed that particulate matter (PM_{10}, $PM_{2.5}$), and O_3 are one the most frequently examined variable in estimating O_3 concentration. Moreover, in a city with a Mediterranean climate like Seville, characterized by warm or hot temperatures for most of the year (with a mean

temperature of 20 °C), the temperature (tmp) also emerges as an important feature. Additionally, given that the data are from a metropolitan area with heavy traffic, the presence of CO_2 is also notable and significantly impacts O_3 concentration. Finally, the importance of wind direction (dd) may be attributed to the presence of both a large urban area and nearby countryside, thus assuming that if the wind is coming from the North-Est, air pollution will be higher, while when coming from the South and West, the air would be cleaner. On the other hand, Fig. 3b depicted the feature importance across the time windows, namely 39, given that measurements from the preceding 39 h have been utilized as input, thus capturing the temporal dimension of each variable. The peak importance values, particularly for CO_2, are concentrated between time windows 10 and 15, and these time intervals approximately correspond to the 24 h preceding the predicted O_3 concentration, indicating a high correlation between the current value and the same time interval from the previous day. Moreover, in the most recent time intervals (from 30 to 39 time window items), the most significant values are the wind velocity (vv) and PM_{10}, features that could represent the current contamination levels: the presence of high wind velocities indicates that strong winds could significantly alter ground-level O_3 concentration.

5 Conclusions and Future Works

Accurate O_3 forecasts are invaluable for effective pollution reduction efforts, and the acquisition of timely and precise information about O_3 levels can guide targeted interventions including emission reduction strategies, traffic management measures, and public health advisories, aimed at minimizing health risks and environmental impact associated with elevated O_3 levels. Besides, ML solutions have already demonstrated their effectiveness in ground-level O_3 concentration forecasting. Several models were tested in this study, and the best results were achieved with an Extra Trees model regarding evaluation metrics and training time. The explanations provided by the RULEx XAI technique illustrate that the model prioritizes features that align with the expertise of domain experts in the literature, namely the concentration of O_3 in past moments, the concentration of CO_2, PM_{10}, air temperature and wind direction. Finally, looking ahead, future research could explore the integration of predictive models with additional datasets, such as a wider range of meteorological variables or real-time satellite imagery, to enhance forecast accuracy and comprehensiveness.

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